

Neoadjuvant/Adjuvant Pembrolizumab and Surgery + Cesium-131 in Recurrent HNSCC: A Multicenter Phase I/IIb Trial.

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Background: Adjuvant external beam re-irradiation after salvage surgical resection for recurrent head and neck squamous cell carcinoma (HNSCC) is associated with significant morbidity without improved survival data. Intraoperative low-dose radiation (LDR) brachytherapy using Cesium-131 (Cs-131) has emerged as a novel method for delivering re-irradiation more safely. Here, we explore the safety and oncologic benefit of Cs-131 brachytherapy combined with neoadjuvant and adjuvant PD-1 inhibition in patients undergoing salvage surgical resection for locally advanced recurrent HNSCC.

Methods: We conducted a multi-institutional, single-arm, non-randomized, open-label phase 1b/II trial. Patients received neoadjuvant pembrolizumab, followed by oncologic resection and intraoperative placement of Cs-131 seeds (delivering 60-70 Gy to the operative bed), and 8 cycles of adjuvant pembrolizumab delivered every 3 weeks. Primary objectives included safety and efficacy, assessed via adverse events (AEs) and disease-free survival (DFS), respectively.

Results: From June 2021 to March 2023, 16 subjects received neoadjuvant immunotherapy (IO), surgery + Cs-131, and one dose of adjuvant IO; 10 completed the full 8 doses. Six patients withdrew prior to completing treatment due to disease progression. 82 treatment related AEs were recorded, with the majority grade 1 or 2 (85% n = 70). There were two grade 4 AEs: one for sepsis, and another for a postoperative oral hemorrhage requiring return to the operating room (OR) for an additional free flap. There were no grade 5 AEs. No AEs led to study treatment discontinuation. The most common treatment related AEs were fatigue (n = 15, 94%), wound dehiscence (n = 6, 38%), and rash (n = 5, 31%). Of the six patients with wound complications, five were in the Cs-131/surgical bed, three of which required return to the OR for flap coverage (grade 3). Six patients experienced disease progression leading to study withdrawal, five within the Cs-131/surgical field, and one in the contralateral neck. A total of 11 patients were evaluable for efficacy. Two-year DFS was 43.6% (95% CI: 21.8–87.4%). DFS showed a steep decline within the first six months postoperatively: 63.6% at 1 month (95% CI: 40.7–99.5%), decreasing to 54.5% (95% CI: 31.8–93.6%) by 6 months, and 43.6% (95% CI: 21.8–87.4%) by 14 months.

Conclusions: This study establishes proof-of-concept showing potential efficacy of neoadjuvant/adjuvant pembrolizumab and surgery + Cs-131, with an acceptable safety profile in patients with recurrent locally advanced HNSCC. Appropriate flap coverage of cesium seeds is imperative to reduce wound complications. Further correlative studies are ongoing to analyze pre- and post-treatment peripheral blood and tumor immune phenotypes and their correlations with outcomes.

Title:

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No

Is this abstract a clinical trial?

Yes

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Yes

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Are patients still being accrued to the trial reported in this abstract?

No

Would like to be considered for a Merit Award:

No

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No

Has this research been submitted for publication in a medical journal?

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